

31.4 A 65nm 1Mb Nonvolatile Computing-in-Memory ReRAM Macro with Sub-16ns Multiply-and-Accumulate for Binary DNN AI Edge Processors

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Many artificial intelligence (AI) edge devices use nonvolatile memory (NVM) to store the weights for the neural network (trained off-line on an AI server), and require low-energy and fast I/O accesses. The deep neural networks (DNN) used by AI processors [1,2] commonly require p-layers of a convolutional neural network (CNN) and q-layers of a fully-connected network (FCN). Current DNN processors that use a conventional (von-Neumann) memory structure are limited by high access latencies, I/O energy consumption, and hardware costs. Large working data sets result in heavy accesses across the memory hierarchy, moreover large amounts of intermediate data are also generated due to the large number of multiply-and-accumulate (MAC) operations for both CNN and FCN. Even when binary-based DNN [3] are used, the required CNN and FCN operations result in a major memory I/O bottleneck for AI edge devices.

A compute in memory (CIM) approach, especially when paired with high-density NVM (nvCIM), can enhance DNN operations for AI edge processors [4,5] by, (1) avoiding latencies due to data access from multi-layer memory hierarchies (NVM-DRAM-SRAM) by storing most, or all, of the weights in the nvCIM; (2) reducing intermediate data access; (3) shortening the latency of multiple MAC operations to one CIM-cycle. As will be discussed later, many challenges and restrictions to circuit designs and NVM devices prevent the realization of nvCIM. Thus far, only one 32x32 CIM ReRAM macro, with a high resistance-ratio (R-ratio) has been demonstrated [6]. Moreover, the 1T1R cell-read currents (I_{LRS} and I_{HRS}), for both the high- (HRS, R_{HRS}) and low-resistance-states (LRS, R_{LRS}), have the same polarity. This prevents nvCIM from implementing both positive and negative weights on the same BL, as is required for software-driven DNN structures, such as XNOR nets. To achieve a smaller energy-hardware cost, this work proposes a hardware-driven binary-input ternary-weighted (BITW) network using our pseudo-binary nvCIM macros and a two-macro (nvCIM-P and nvCIM-N) DNN structure [5]: the nvCIM-P macro stores positive weights while the nvCIM-N stores negative weights. The BITW network combines ternary weights [4] (+1, 0 and -1) and a modified binary input (1 and 0) with an in-house computing flow for subtraction (S), activation (A) and max-pooling (MP). This work proposes various circuit-level techniques for the design of a 65nm 1Mb pseudo-binary nvCIM ReRAM macro capable of storing 512K weights, and performing up to 8K MAC operations within one CIM cycle. We demonstrate the first megabit nvCIM ReRAM macro for CNN/FCN, and the fastest (<16ns) CIM operation for NVMs.

Figure 31.4.2 shows the structure of our BITW nvCIM 1T1R ReRAM macro: comprising of a dual-mode WL driver (D-WLDR), a distance-racing current-mode sense amplifier (DR-CSA), a reference current (I_{REF}) generator (REFG), and 1T1R ReRAM cell arrays. All binary weights (W) of each $n \times n$ kernel in a CNN layer or m weights in a FCN layer are stored in memory cells on the same BL. A 0-weight is encoded as HRS, while a +1 weight is encoded as LRS in nvCIM-P, and a -1 as LRS in nvCIM-N. Digital inputs are applied via WL. When WL=0 the cell current (I_c) of a cell is zero. When WL=1, I_c is either I_{HRS} (for 0-weight) or I_{LRS} (+1 or -1-weight). If the cell's R-ratio is large ($I_{HRS} \ll I_{LRS}$), then I_c is equal to the product of input $\times$ weight. Using a current-mode read scheme, the action of turning on [0, n^2 (or m)] WL simultaneously results in an I_{BL} that is equal to summation of all activated cells (I_c). MAC values (MACV) which refer to the number of $IN \times W = \pm 1$ are differentiated using a multi-level (ML) CSA to resolve I_{BL} and output a j -bit digital code. The j values are determined by an algorithm in an NN model structure. The proposed nvCIM with v -columns and s -IO can perform up to $(s \times n^2$ or $s \times m)$ MAC operations within one CIM cycle, where $s \leq v$.

There are three major challenges in using a nvCIM ReRAM macro. (1) Large input offset (I_{OS}) for the CSA when I_{BL} is large. (2) A reduced sensing margin (SM), when using a conventional mid-point I_{REF} read scheme. (3) A small sense margin across various computing (MACV) states due to input-patterns and process variation; particularly when the R-ratio is small (significant I_{HRS}). We propose a DR-CSA to suppress I_{OS} at high I_{BL} while efficiently utilizing the signal margin in order to overcome (1) and (2). Furthermore, we proposed an input-aware dynamic I_{REF} generation scheme (IA-REF) to combat (3).

Figure 31.4.3 presents DR-CSA, comparing the distance between I_{BL} and two I_{REF} ($I_{REF,H}$, $I_{REF,L}$), using the top (PT1, PT2, PT3) and bottom (PB1, PB2, PB3) current paths. In phase-1, P0-P2 precharges the BL and reference-BLs (RBL). S1 is on to store the gate-source voltage of P0 ($V_{GS,P0}$) and P1 ($V_{GS,P1}$) on C0/C1 to sample the $I_{BL}/I_{REF,H}$ (at

PT2/PT3) as in [7]. In phase-2, PRE/S1 is off and S2/S3 are on. Path PT2 (I_{BL}) is connected to PB1 ($I_{REF,L}$), while PT3 ($I_{REF,H}$) is connected to PB2 (I_{BL}). For a given period T_{P2} , the voltage at node Q (V_Q) is $T_{P2}(I_{BL}-I_{REF,L})$, while the voltage at node QB (V_{QB}) is $T_{P2}(I_{REF,H}-I_{BL})$. The difference between V_Q and V_{QB} ($\Delta V_{Q-QB} = |V_Q - V_{QB}|$) is $(2I_{BL} - (I_{REF,H} + I_{REF,L}))T_{P2}$. In phase-3, S2 is off, and the cross-couple pair N3/N4 amplifies ΔV_{Q-QB} as the phase-3 period (T_{P3}) increases. Finally in phase-4 SAEN is on to generate a digital output (SAO) based on ΔV_{Q-QB} . In short, the sensing margin of DR-CSA is $(2I_{BL} - (I_{REF,H} + I_{REF,L}))$, which is approximately $2 \times$ larger than a conventional mid-point CSA ($I_{BL} - (I_H + I_L)/2$). Repeating this sensing procedure j times generates a j -bit output by DR-CSA. Many BLs share one CSA; therefore, the area overhead of DR-CSA is less than 1.54% for a 1Mb macro.

Figure 31.4.4 presents the IA-REF scheme comprising of a reference-WL controller (RWLC), input-counter (ICN), and input-aware replica rows (IA-RR). A limited R-ratio and significant I_{HRS} leads to overlap in I_{BL} across neighboring MACV values, resulting in read failures. For example, the I_{BL} for MACV=1 (one LRS cell) may be due to one WL on (one LRS cell, 1L0H) or nine WLs on (one LRS and 8 HRS cells, 1L8H). The $I_{BL,1L8H}$ is larger than the minimum I_{BL} for MACV=2 (two LRS cells accessed by two WLs, 2L0H). Fortunately, for MACV with the same number (N_{WL}) of WL on, there is spacing in I_{BL} between neighboring MACVs.

The top and bottom sub-arrays in this nvCIM macro operate as a pair. When accessing the top array, selected BLs in the bottom array serve as reference BLs (RBL). For a nvCIM with j -bit ML outputs, 2^j BLs per IO are used as RBLs. The input-counter counts the number inputs equaling 1 in the input pattern, and outputs N_{WL} to RWLC. Then, RWLC turns on N_{WL} replica WLs (RWLs), from RWL[1] to RWL[N_{WL}]. IA-RR includes k rows of RWLs with designed patterns on their replica memory cells (RMC) to generate various I_{REF} across RBLs for DR-CSA. A RBL[r] has r LRS cells, where the range of m is [0, ($2^j - 1$)]. Each ML sensing cycle activates only two RBLs for the required $I_{REF,H}$ and $I_{REF,L}$.

For example, there are 9 RWLs and 8 RBLs in a IA-RR for a CNN operation with 3×3 kernels (max. 9 inputs) using 3-bit ML DR-CSA ($j=3$). If an ML sensing cycle requires $I_{REF,L}=3L5H$ and $I_{REF,H}=4L4H$ at $N_{WL}=8$, then RWL[1]-RWL[8] are on and RBL[3] and RBL[4] are activated. RBL[3] has 3 LRS cells (RMC[3:1,3]) and 5 HRS cells (RMC[8:4,3]) activated to generate $I_{RBL3}=3I_{LRS}+5I_{HRS}$. RBL[4] has 4 LRS cells (RMC[4:1,4]) and 4 HRS cells (RMC[8:5,4]) activated to generate $I_{RBL4}=4I_{LRS}+4I_{HRS}$. Figure 31.4.5 shows that with a larger sensing margin and current-distance scheme, the proposed DR-CSA enables a 5-6 $\times$ reduction in input offset, compared to a conventional CSA with an I_{BL} exceeding 40uA. With the same sensing yield across MACVs, DR-CSA can tolerate a minimum R-ratio that is $4 \times$ smaller than a conventional CSA. The IA-REF scheme increased the worst-case signal margin of MACVs from -27.9uA to 7.8uA. DR-CSA and IA-REF enhance the accuracy of text recognition/inference operations (i.e. MNIST database) by 50 $\times$ for binary DNN, compared to a conventional CIM (not using DR-CSA and IA-REF).

Figure 31.4.6 presents the measured results of a 1Mb nvCIM ReRAM macro fabricated using 1T1R contact-ReRAM cell in a 65nm CMOS logic process. For CNN operations using 3×3 kernels the measured CIM access time (T_{AC-CIM}) excluding the path-delay (T_{PATH}) is 14.8ns. In FCN mode, the shmoo plot confirmed a $T_{AC-CIM}=15.6ns$ across various input patterns with 25 inputs per operation. This work increased the computing speed by $2 \times$ and capacity by 1000 $\times$, compared to previous ReRAM nvCIM [6]. Figure 31.4.7 presents a die photo and chip summary.

Acknowledgements:

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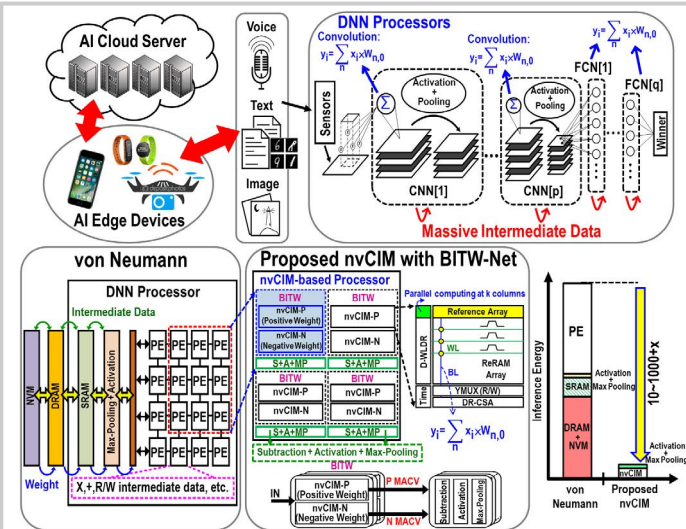


Figure 31.4.1: Nonvolatile computation in memory (nvCIM) concept, with BITW-NN in AI edge processors.

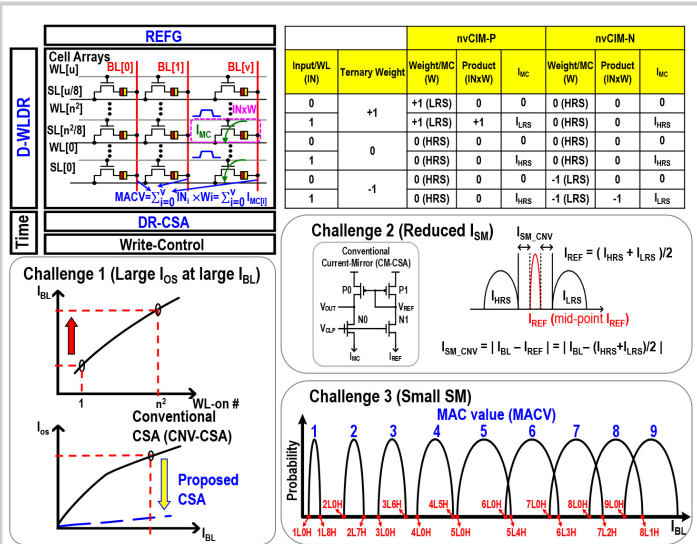


Figure 31.4.2: Proposed macro structure of nvCIM and its challenges.

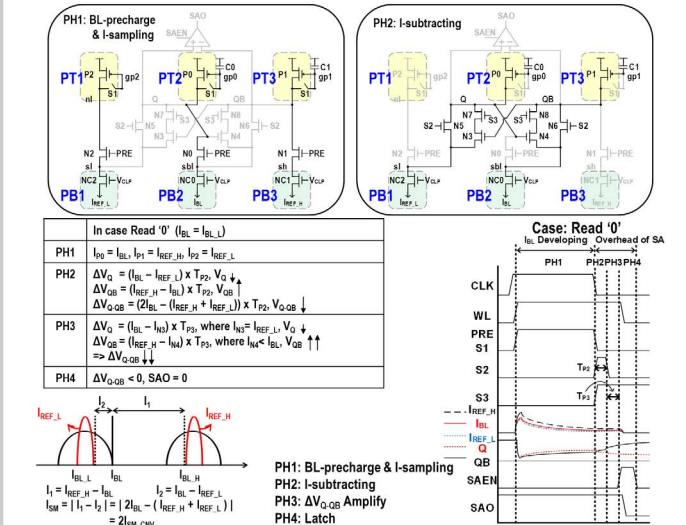


Figure 31.4.3: Structure and operation of distance-racing current-sense amplifier.

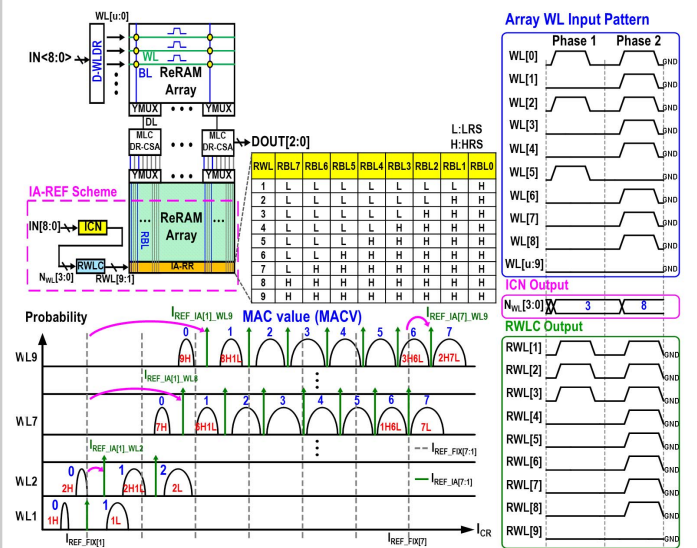


Figure 31.4.4: Proposed input-aware dynamic I_{REF} generation scheme.

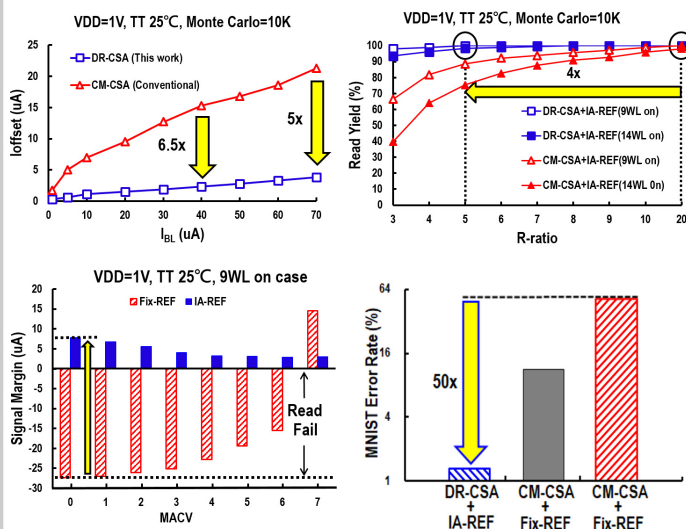


Figure 31.4.5: Performance of proposed schemes.

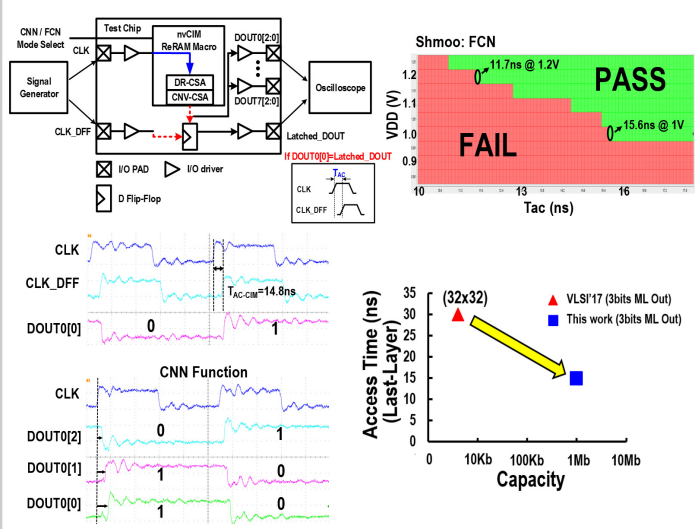
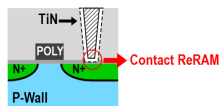
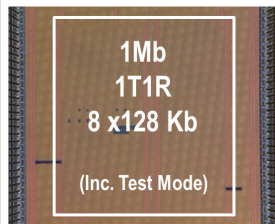


Figure 31.4.6: Measured results.



Technology	65nm CMOS Logic Process
ReRAM	Logic-process compatible CRRAM (Unipolar)
Cell size(1T1R)	0.25 μm^2
RRAM mode	Memory/ CIM
CIM mode	CNN/ FCN
Capacity	1Mb (8 x 128 Kb)
Sub-array	512 rows x 256 columns
Read Delay for BL-length=512 @ VDD=1V	Memory mode: 5.0 ns(1bit Out) CNN mode: 14.8 ns(3bits ML Out) FCN mode: 15.6 ns(3bits ML Out)

Figure 31.4.7: Die photo and summary table.